DSA5208 GROUP PROJECT

# Rainfall across Singapore

Patterns and next-30-minute risk, 2017–2024


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**Abstract**

Singapore's five-minute rainfall records offer a detailed view of local weather, while uneven coverage affects comparisons over time. We analyse the 2017–2024 annual datasets with Apache Spark, retaining 48.54 million observations from 91 station IDs after cleaning. Among sufficiently complete station-months, November has the highest mean monthly rainfall and wet intervals are most frequent in the afternoon. For prediction, we compare random forests, gradient-boosted trees, logistic regression and historical frequencies under two information settings: recent station rainfall and station/calendar information alone. On 5.73 million shared 2024 test cases, the recent-rainfall forest achieves mean Brier 0.01613, compared with 0.02283 for the calendar forest. Random forests give the lowest mean probability error among the three classifiers; historical frequency performs best for calendar-only risk. Increasing fitting samples from 15% to 100% produces small changes in the reported scores. Gradient boosting ranks 1 mm events competitively but overpredicts the average risk of large accumulations. The results favour recent-rainfall forests for short-term probability queries and identify high-threshold calibration as a useful direction for further modelling.


## 1 Introduction

The project asks two related questions: how rainfall varies across Singapore within and between years, and how well we can estimate a station's chance of receiving more than a specified amount in the next 30 minutes. The first question concerns patterns across many observations. The second requires a clear account of what information would be available when a forecast is issued.

We use all eight annual datasets from 2017 to 2024. For exploration, the main difficulty is uneven coverage: a low recorded monthly total can reflect missing observations as well as dry weather. We address this through completeness rules and comparisons at common stations. For prediction, we distinguish a calendar-only model from a model that also uses recently published rainfall. This lets us examine both the three-input query in the assignment—station, prediction time and threshold—and the benefit of supplying current weather observations internally.

The report follows these two questions from data preparation to results. Spark handles the observation-level calculations and model fitting; figures and supplementary analyses use the saved result tables.

## 2 Data and methods

### 2.1 What the records contain

We use the eight annual Historical Rainfall across Singapore datasets for 2017–2024. Each observation contains a station identifier, coordinates, a timestamp, rainfall in millimetres and update metadata. All times are interpreted in Singapore time. The measurement is an amount accumulated over a five-minute interval, rather than a rainfall rate. The publisher cautions that records may be missing and have not undergone the quality control applied to official climate records [1].

The cleaned dataset contains 48,538,625 observations from 91 station IDs. Recorded zeros account for 97.08% of observations, while positive amounts have a long upper tail (Figure 1). We therefore examine how often rain occurs separately from how much falls during wet intervals. A zero is a measurement of no rain; a missing record provides no such information. Appendix A provides the positive-rainfall distribution and a raw-versus-clean source-gap example.

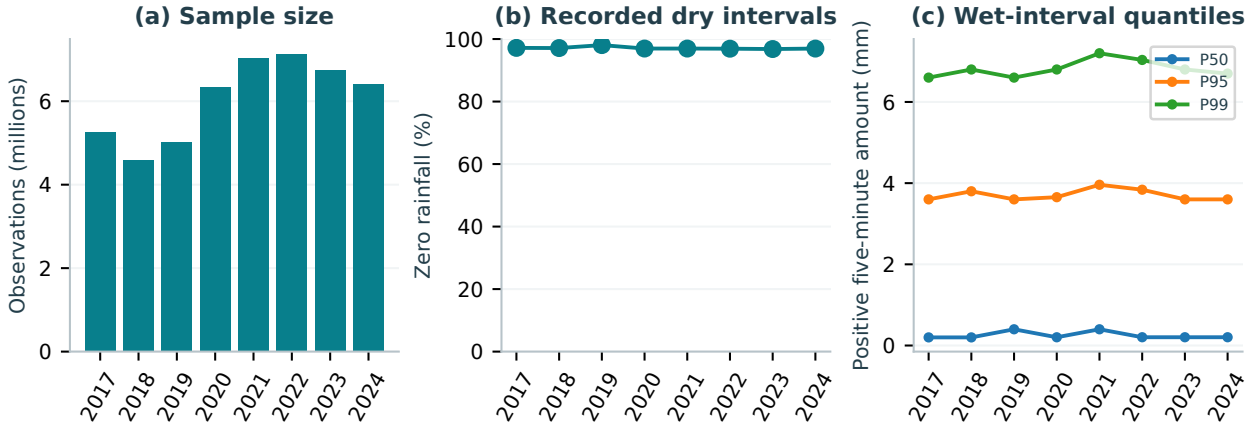



**Figure 1.** The observation sample by year: record counts, the percentage of recorded zeros, and quantiles of positive five-minute rainfall. Positive-rainfall quantiles exclude dry intervals.

## 2.2 Cleaning and missing observations

The PySpark pipeline standardises identifiers and field types, checks rainfall units and measurement type, and rejects malformed records, invalid timestamps, invalid coordinates, and non-finite or negative rainfall. Timestamps within two seconds of a five-minute boundary are rounded to that boundary; records farther from the grid are excluded. Exact duplicates are removed. If records sharing a station and time disagree on rainfall or coordinates, or include an invalid source record, all records for that station and time are set aside for inspection. Consistent measurements with different update metadata are consolidated using the latest publication timestamps, conservatively delaying their availability for prediction.

The source-to-clean audit reconciles 48,538,713 raw rows with 48,538,625 clean observations. The count falls by 88; because cleaning also consolidates repeated records, this difference is not a count of invalid measurements. All raw timestamps were parseable. High non-negative amounts are retained for review, including the maximum of 37.2 mm in five minutes; the cleaning rules alone do not establish whether an extreme reading is meteorologically valid. The pipeline preserves the raw files and writes rejected records, quality summaries and clean Parquet partitioned by year and month. The repository includes the summary evidence; the full raw and intermediate datasets require reproduction through the documented cloud workflow.

Missingness matters more to this analysis than the small change in row count. The source audit identifies 63 calendar dates without observations anywhere in the network. August 2018, for example, contains 36,410 raw records, all within its first three days. This gap was already present in the annual source file. Timestamp rounding can also move near-midnight records to an adjacent date, so daily raw-to-clean differences need not represent deletions.

For station $s$, year $y$ and month $m$, we measure completeness as

$$C_{sym} = \frac{N_{sym}}{288 D_{ym}},$$

where $N_{sym}$ is the number of valid unique observations and $D_{ym}$ is the number of calendar days. Every station is represented in all 96 year-months; absent months have zero observations and an undefined rainfall total. Low coverage can consequently reflect a station's operating period as well as missing records.

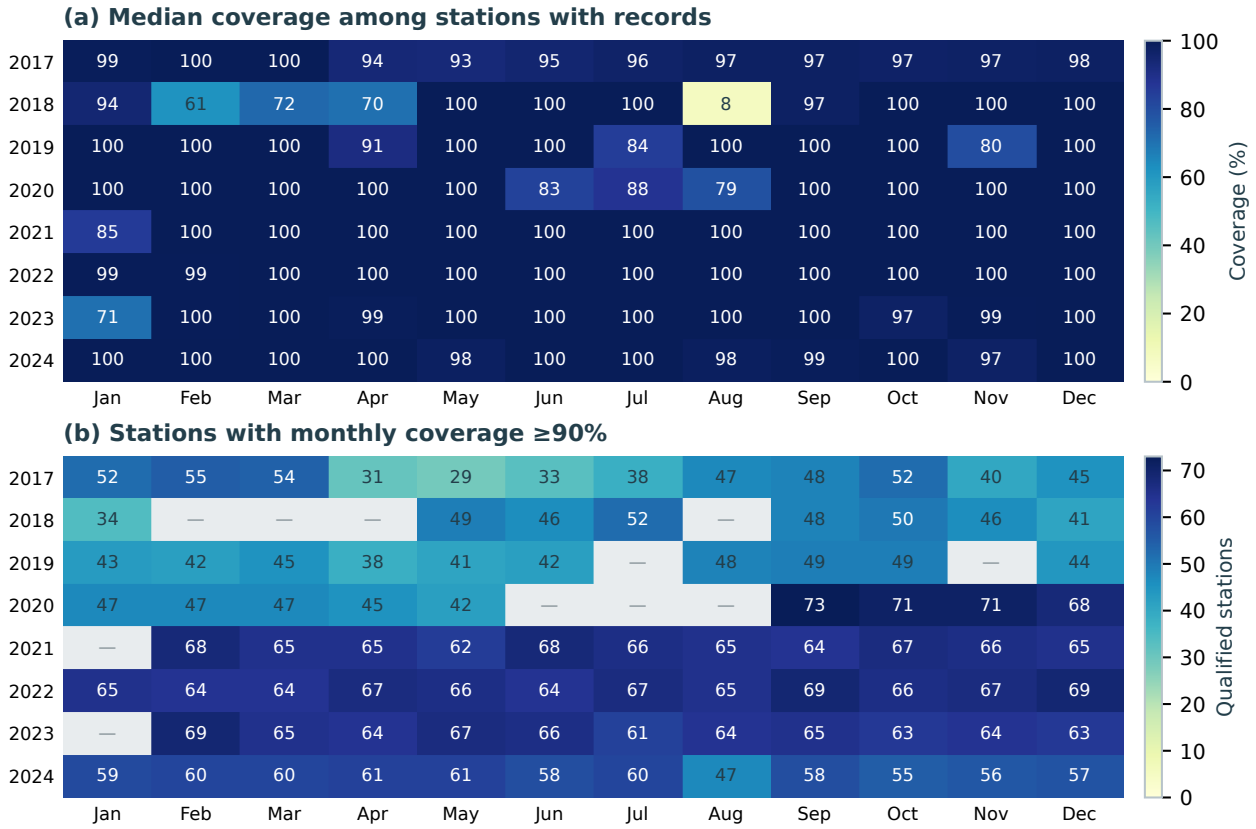



**Figure 2.** Monthly data availability. The upper panel gives median coverage among stations with any observations; the lower panel counts stations meeting the 90% rule. Grey cells in the lower panel have no qualifying stations.

## 2.3 Comparing rainfall over time

Monthly summaries require at least 90% coverage. Annual summaries require 95% coverage overall and 90% in every month, preventing a concentrated seasonal gap from being hidden by a high annual average. Daily-distribution summaries require 95% daily coverage within eligible months. These rules limit missing-data exposure but cannot remove bias if missingness is related to weather. Totals remain observed amounts, without gap filling or scaling by inverse coverage.

Within each year-month, we average eligible station totals equally. Seasonal curves then weight available years equally. Wet frequency is the number of positive intervals divided by observed intervals; conditional intensity is their rainfall sum divided by positive intervals. Hourly summaries pool eligible observations, giving more weight to periods with more observed intervals. Station means describe the sampled network, rather than an area-weighted national rainfall total.

Cross-year comparisons match station IDs within the same calendar month for each year pair. Coordinate stability screens use a tolerance of 0.005 degrees per axis within and between paired periods. We average the within-station changes and express percentage change relative to the earlier pair mean. Plots show only comparisons with at least five common stations; this is a display threshold, not a significance test. Different year pairs can contain different stations.

## 2.4 Computing environment

Observation-level processing, feature construction and Spark MLlib training ran on Google Cloud Dataproc. Table 1 records the experimental configuration. Only saved summaries were downloaded for local analysis and plotting; source snapshots and job metadata accompany the results.

| Component | Configuration |
|---|---|
| Cluster | One master and two workers in `asia-southeast1-c` |
| Each VM | `e2-standard-4`: 4 vCPUs, 16 GB RAM, 80 GB `pd-balanced` boot disk |
| Worker capacity | 8 vCPUs and 32 GB RAM, excluding the master |
| Runtime | Dataproc 2.2.87, Debian 12; Spark/PySpark 3.5.3 on YARN |
| Spark settings | 6 GB driver memory; 1 GB maximum driver result; 128 shuffle partitions; adaptive SQL; Singapore timezone |
| Local figures | Apple M2, 16 GB RAM; pandas, Matplotlib and Plotly |
| Storage | Cloud Storage for raw files, Parquet data and result tables |

**Table 1.** Configuration used for the reported experiments.

## 3 Rainfall patterns, 2017–2024

### 3.1 Seasonality and changes between years

The clearest seasonal feature is the increase in rainfall towards the end of the year. November has the highest mean observed monthly amount, 320.8 mm, while February has the lowest, 146.1 mm. These averages give equal weight to each available year after averaging its qualified stations. Figure 3 shows the individual year-months behind this summary; Figure 4 separates rainfall amount into frequency and intensity. The broad pattern agrees with the wetter November–January period and relatively dry February described by the Meteorological Service Singapore [2].

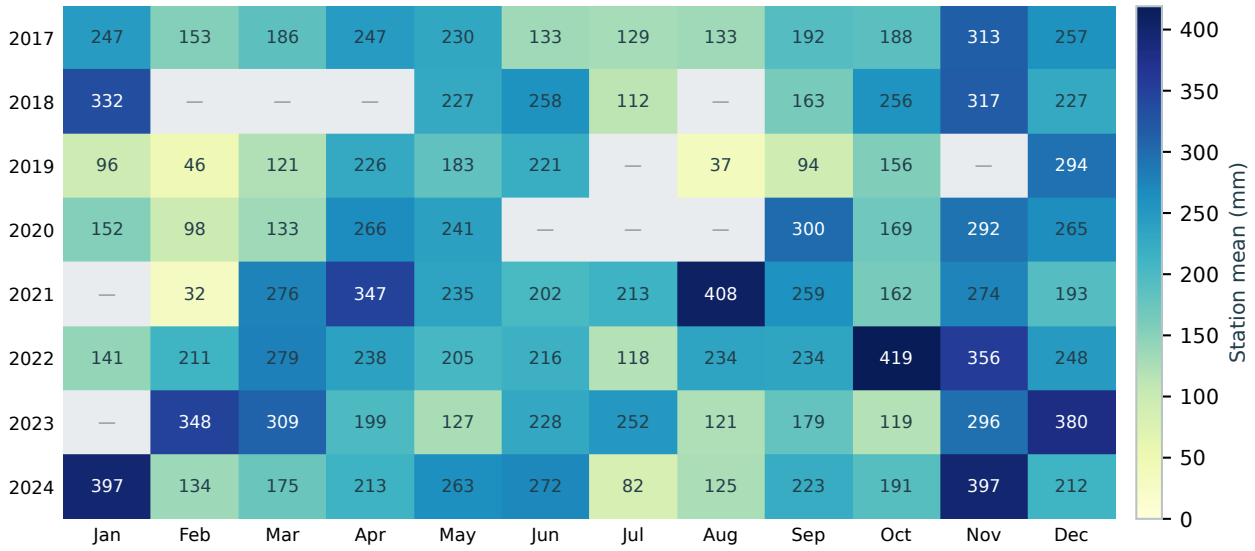



**Figure 3.** Mean observed monthly rainfall at qualified stations, 2017–2024. All cells share one colour scale. Grey cells have no qualified station-months.

The frequency and intensity curves help explain what makes a month wet. November has the highest mean wet-interval frequency, approximately 4.30%, but its mean amount per wet interval is about 0.89 mm. March has a higher conditional intensity, about 1.06 mm, despite a lower monthly total. Thus, November's high accumulation is associated with more frequent rain, rather than the largest amount in each wet interval. This distinction matters for prediction: a season with frequent rain need not have the greatest risk of a large amount within a short window.

Monthly totals also reflect the number of days in a month. February's shorter duration contributes to its low total, so the amount curve alone cannot establish that it has the lowest propensity for rain. The frequency and intensity panels provide complementary measures. Between six and eight years contribute to each calendar-month average, and the shaded ranges describe their observed variation rather than uncertainty in a long-term climate normal.

To examine changes between years, we compare the same calendar month at stations present in both periods

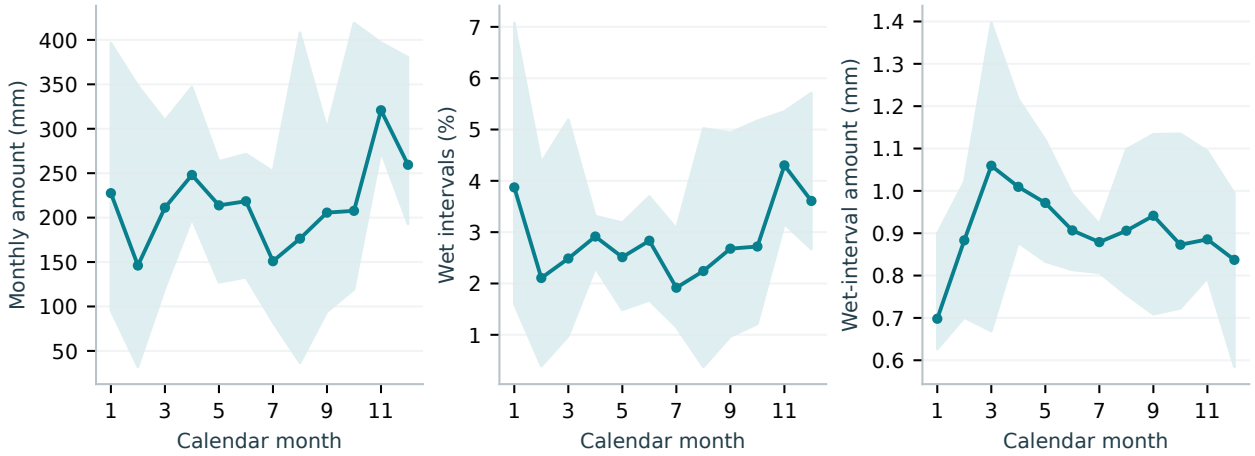



**Figure 4.** Seasonal means of monthly rainfall, wet-interval frequency and rainfall per wet interval. Available years receive equal weight; shading shows their minimum–maximum range.

and meeting the location-stability checks. December 2024, for example, had a mean of 213.9 mm across 53 common stations, compared with 379.8 mm in December 2023: a reduction of 43.7%. This provides stronger evidence of a change at those locations than a comparison of two unmatched network means. Figure 5 applies the same approach to adjacent-year monthly pairs. The participating stations vary between pairs, so the percentages describe separate comparisons and cannot be chained into a national trend.

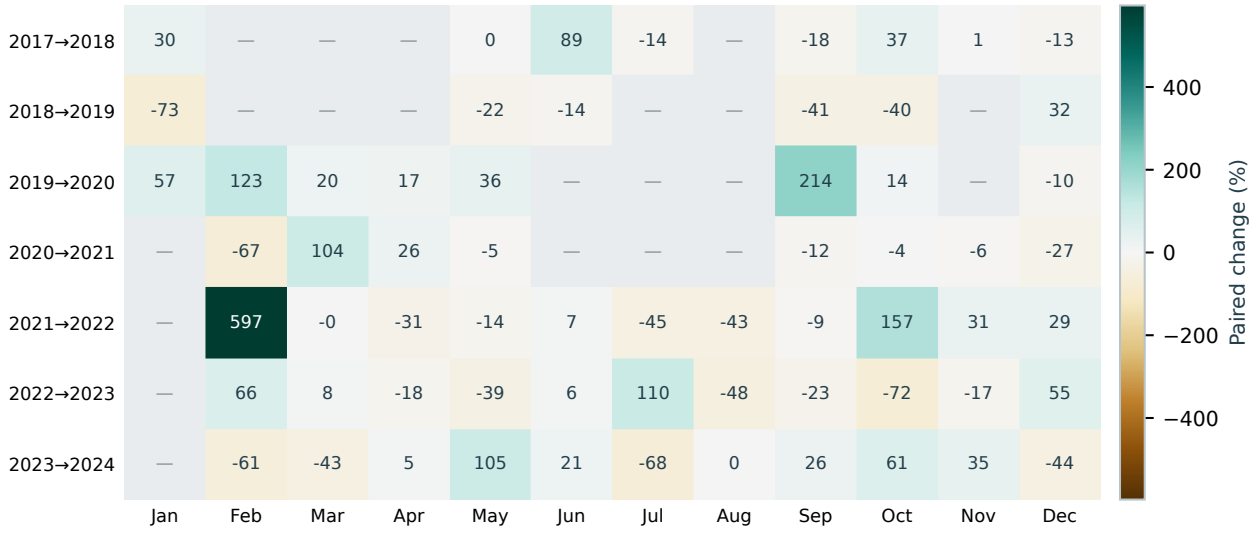



**Figure 5.** Adjacent-year changes in monthly rainfall at common stable-location stations, relative to the earlier year's mean. Grey cells lack support from at least five common stations; this cutoff is a display rule, not a significance test.

Full-year comparisons are more limited. Only 2017, 2022 and 2024 contain stations meeting both the annual and every-month completeness requirements. Their mean observed annual totals are 2,396.6 mm (9 stations), 2,920.9 mm (54 stations) and 2,682.3 mm (29 stations), respectively. Among the 26 stations eligible in both 2022 and 2024, the mean declined by 284.8 mm, or 9.6%. Comparisons involving 2017 have only three or four common stations. These results support a limited comparison of monitored locations, but neither an eight-year annual trend nor a national wettest-year ranking. Months with adequate coverage in the other years remain part of the monthly analysis.

## 3.2 When and where rain occurs

Rain is more frequent in the afternoon. Across valid intervals in qualified station-months, wet frequency reaches 5.80% at 15:00–15:59 and falls to 1.10% at 23:00–23:59, using Singapore time. Figure 6 shows how this daily pattern varies by month and year. The afternoon enhancement is consistent with MSS's description

of Singapore's rainfall [2] and motivates including hour-of-day features in the predictive model. These pooled frequencies describe observed occurrence; they are not forecasts for a particular afternoon, and the yearly curves can contain different mixes of months and stations.

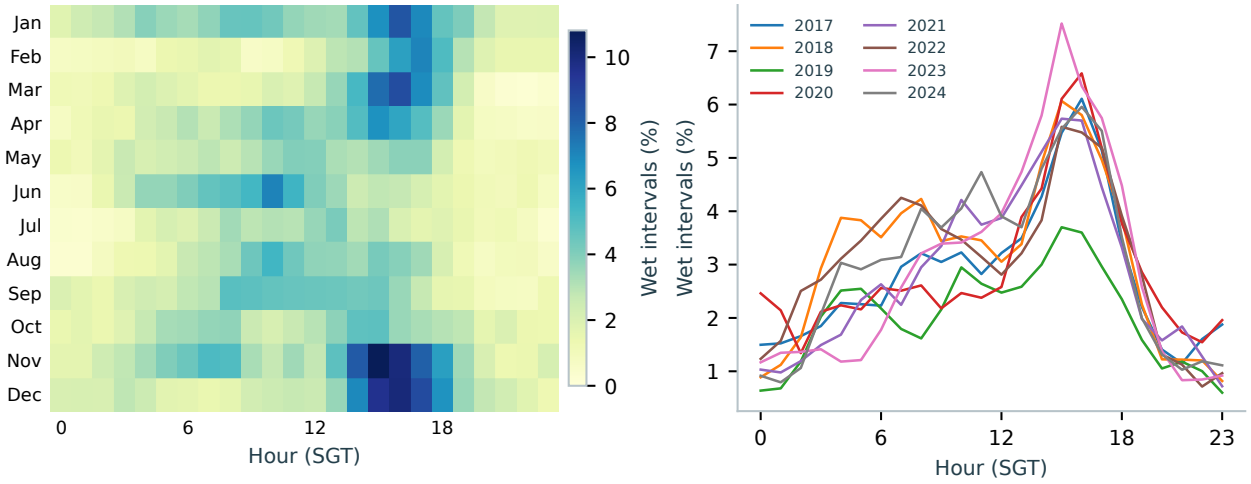



**Figure 6.** Wet-observation frequency by month and hour, and by year and hour, in Singapore time. Frequencies are weighted by the number of observed intervals.

Rainfall also varies substantially across monitoring locations. In December 2024, qualified station totals range from 29.5 mm at S224 to 326.8 mm at S117. The December maps retain 41–69 stations in each of the eight years, making this month a useful illustration of spatial and year-to-year variation. Figure 7 shows the measured station values on a common scale. The full set of 96 year-month maps remains in the project repository's supplementary gallery. The map outline comes from the retained boundary assets [3]. We avoid interpolating a continuous surface because the station network alone does not establish how rainfall varies between gauges; station means likewise should not be interpreted as area-weighted national rainfall.

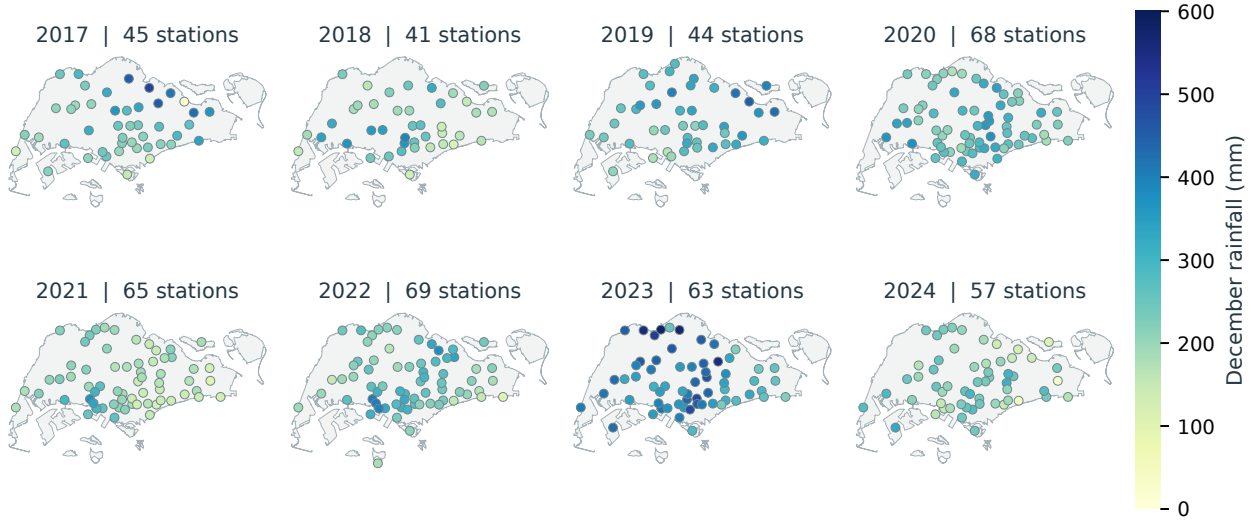



**Figure 7.** December rainfall at qualified stations in each study year. All eight panels use a 0–600 mm scale, covering the full range of December station totals. Panel labels give station counts; colours represent measurements at gauges, with no spatial interpolation.

### 3.3 How sensitive are these summaries to coverage?

Changing the monthly coverage cutoff produces relatively small changes in the supported monthly means. Relative to the 90% cutoff, using 80% gives a median absolute change of 0.61% across 85 shared year-months, with a maximum of 5.78%. Using 95% gives a median change of 0.40% across 84 shared year-months, with

a maximum of 2.97%. This is reassuring for the descriptive amount summaries, although each cutoff changes the contributing stations. It does not establish that missing observations are random. Residual gaps, uneven station coverage and the short study period remain the main reasons to treat these findings as patterns in the available observations rather than evidence of long-term climate change.

Appendices B–D retain monthly trajectories, annual station totals, daily-rainfall summaries, coverage-sensitivity curves and additional spatial views. They supplement the principal comparisons without requiring all stations to qualify in all eight years.

## 4 Predicting the next 30 minutes

### 4.1 What information is available when a prediction is made?

For station $s$, issue time $t$ and threshold $q$ in millimetres, the target is

$$Y_{s,t} = \sum_{j=1}^{6} r_{s,t+5j\,\mathrm{min}}, \qquad p_{s,t}(q) = P(Y_{s,t} > q \mid X_{s,t}).$$

We assume that observation timestamps mark the endpoints of five-minute accumulation intervals. A training label requires all six exact future records; taking the next six available rows could incorrectly span a gap. Totals are rounded to 0.001 mm, and exceedance means strictly greater than $q$. Eligibility depends on these local windows, rather than Task 1's monthly coverage filter.

We consider two information settings for this query. A uses recent rainfall from the requested station to estimate short-term risk. B uses station and calendar information to estimate historical risk at the requested time. Our primary focus is A: recent observations provide additional information about current rainfall conditions and, as the results show, improve predictive performance. B provides a baseline without recent rainfall and can also estimate risk for future times whose recent observations are not yet available, such as tomorrow afternoon. This estimate reflects historical station and calendar patterns rather than the weather conditions of a particular future storm. The same model families are tested in both settings, allowing us to examine how much the available information matters alongside the choice of algorithm.

**A, the short-term model**, also reads recent rainfall from a supplied clean observation table. We use a prespecified ten-minute buffer, with twelve observations ending at $t-65, t-60, \ldots, t-10$ minutes. Each record must be present and both recorded update timestamps must be no later than $t$. Missing or not-yet-updated history excludes an example instead of becoming zero rainfall.

Table 2 helps explain this choice. The delay between observation and the later of the two update timestamps has a median close to ten minutes. We combine the buffer with record-level checks to account for records with longer update delays.

| Period | Observations | Median (min) | P90 (min) | Within 10 min (%) |
|---|---|---|---|---|
| 2017–2021 (training) | 28,256,245 | 9.58 | 15.08 | 68.93 |
| 2017–2024 (all years) | 48,538,625 | 9.47 | 12.68 | 76.40 |

**Table 2.** Observation-to-update delay in the cleaned records. All observations have both update timestamps. P90 is the 90th percentile; the final column is the percentage with the later update no more than ten minutes after observation. These counts describe observations, not eligible prediction examples.

Figure 8 shows the separation between predictors and the future label.

**B, the calendar model**, needs only station, time and threshold. It estimates the historical risk associated with that place and time of year or day. For example, B can answer a query about tomorrow afternoon; A can use tomorrow's recent observations only when they become available.

Both models include station ID, latitude, longitude, and sine/cosine encodings of time of day and annual phase. The annual encoding accounts for leap years; numeric year is excluded. A adds the latest three five-minute

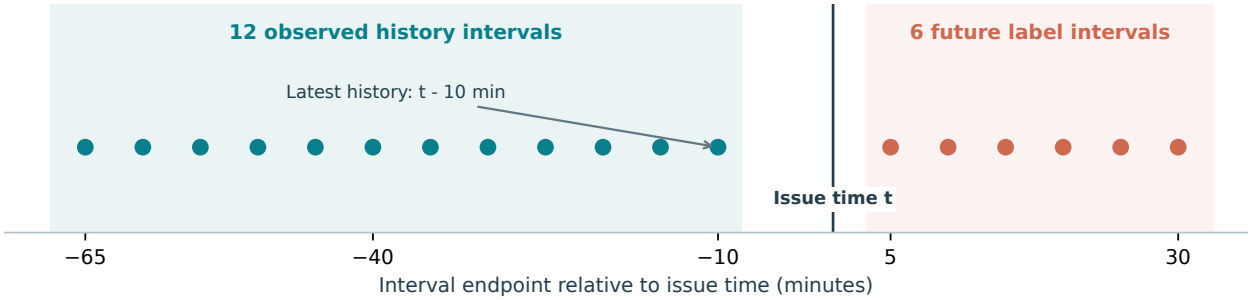



**Figure 8.** The information used by A at issue time $t$. Markers show interval endpoints. The latest history interval ends ten minutes before issue time; the six intervals after issue time form the target.

amounts, 15-, 30- and 60-minute totals, the 30-minute maximum, the hour's wet count and dry fraction, and the difference between consecutive 15-minute totals. The rainfall summaries represent recent intensity, persistence and change. Cyclic time features capture the seasonal and daily patterns without introducing a discontinuity at midnight or New Year. This gives A sixteen numeric features and B six, each alongside station ID.

Table 3 lists the inputs in each setting. All rainfall windows end at $t - 10$ minutes.

| Feature group (number of numeric features) | A | B |
|---|---|---|
| Latitude and longitude (2) | Yes | Yes |
| Time of day: sine and cosine (2) | Yes | Yes |
| Annual phase: sine and cosine (2) | Yes | Yes |
| Latest three five-minute rainfall amounts (3) | Yes | No |
| Rainfall totals over 15, 30 and 60 minutes (3) | Yes | No |
| Maximum five-minute rainfall within 30 minutes (1) | Yes | No |
| Wet-interval count and dry-interval fraction over 60 minutes (2) | Yes | No |
| Difference between the latest and preceding 15-minute totals (1) | Yes | No |
| **Total numeric features** | **16** | **6** |
| Station ID (categorical; counted separately) | Yes | Yes |

**Table 3.** Input features for the two information settings. A uses recent rainfall in addition to the station and calendar inputs shared with B. For LR and GBT, the query threshold $q$ is an additional input; RF uses it when converting class probabilities to exceedance probabilities.

## 4.2 Experimental design

Adjacent examples share much of their history and future rainfall, so a random train/test split would mix closely related observations. We instead assign separate years to training, selection, calibration and testing, removing labels whose target end reaches the following split boundary.

| Period | A examples | B examples | Purpose |
|---|---|---|---|
| 2017–2021 | 18,688,263 | 27,434,001 | Training population |
| 2022 | 6,133,684 | 7,107,898 | Model and cutoff selection |
| 2023 | 5,465,591 | 6,719,274 | RF and GBT temperature selection |
| 2024 | 5,729,508 | 6,374,476 | Model evaluation |

**Table 4.** Eligible station-time examples after completeness checks and boundary purging. Fitting fractions apply only to 2017–2021; validation, calibration and testing use every eligible example.

A has fewer examples because it additionally requires complete, available history. We investigate fitting fractions of 15%, 30% and 100%. Sampling uses a deterministic station–time hash with seed 2026 and does not depend on the label; the smaller samples are contained in the larger ones. The 30% samples contain

5,608,769 examples for A and 8,233,882 for B. Even the 15% samples contain millions of examples, but only 2,886 and 4,198 exceed 20 mm, respectively, and many windows overlap within a rainfall episode.

We compare RF, LR and GBT at each of three fitting fractions: 15%, 30% and 100%, separately in A and B. Every model uses 2017–2021 fitting data and is evaluated on the same complete eligible 2024 population within its information setting. Table 5 summarises the common protocol. Each algorithm, fraction and information setting has its own fitted model; the models are not combined into an ensemble. The RF's within-bin rainfall distributions and the historical-frequency baseline always use the full eligible training population.

| Algorithm | Fitting fractions | 2022 selection | 2023 calibration | 2024 test |
|---|---|---|---|---|
| RF | 15%, 30%, 100% | Reference configuration; alarm cutoff | Temperature | All eligible examples |
| LR | 15%, 30%, 100% | L2 regularisation; alarm cutoff | None | All eligible examples |
| GBT | 15%, 30%, 100% | Tree depth | Temperature | All eligible examples |

**Table 5.** Common model-comparison protocol in both A and B. Fitting fractions apply only to 2017–2021 anchors. RF keeps the reference 40-tree, depth-8 configuration across fractions; the saved 30% forest is calibrated and tested using the same procedure as the other forests. Historical frequency is a full-training comparator.

The RF 30% weights were already available from the earlier validation study. We reuse those standard-configuration forests without refitting and complete their 2023 calibration and 2024 evaluation. The additional one-factor RF tuning analysis remains in Appendix G; it supplements the unified 2024 comparison rather than replacing a missing test cell. Model-file origins and training batches are documented in Appendix F.

Model selection uses equally weighted mean Brier over seven rainfall thresholds: 0, 0.2, 1, 2, 5, 10 and 20 mm. Where calibration is applied, its temperature is selected on 2023 from 0.5, 0.75, 1, 1.25, 1.5 and 2. We also evaluate 3, 7.5 and 15 mm as additional test queries between the training thresholds.

### 4.3 Four approaches to the probability query

**Distributional random forest (RF).** The forest assigns the future total to seven bins: $\{0\}$, $(0, 1]$, $(1, 2]$, $(2, 5]$, $(5, 10]$, $(10, 20]$ and $(20, \infty)$ [4]. We convert its bin probabilities into an exceedance probability by combining them with the observed distribution of training amounts inside each bin:

$$\widehat{p}(q \mid X) = \sum_{k=0}^{6} \widetilde{\pi}_k(X)\widehat{S}_k(q), \qquad \widetilde{\pi}_k = \frac{\pi_k^{1/T}}{\sum_\ell \pi_\ell^{1/T}}.$$

Here $\pi_k(X)$ is the forest probability for bin $k$, and $\widehat{S}_k(q)$ is the fraction of training amounts in that bin exceeding $q$. The temperature $T$ adjusts how concentrated the bin probabilities are; $T = 1$ leaves them unchanged. The resulting exceedance probability is non-increasing in the requested rainfall threshold. The within-bin distributions are shared across weather conditions and extend to the largest training total, 71.8 mm; the model returns zero above this empirical support.

The pipeline indexes station IDs as categorical features and fits numeric median imputation; missing history is still excluded before fitting. The reference configuration has 40 trees, 25 minimum examples per node, square-root feature subsets and 0.8 subsampling. The 15% and 100% searches compare depths 5 and 8 and select depth 8 in both information settings; the 30% forests use this same reference depth. Temperature selection chooses $T = 1$ for all three fractions in both settings, leaving the class probabilities unchanged.

**Threshold-conditioned logistic regression (LR).** LR predicts the binary label $\mathbf{1}(Y > q)$ with the rainfall threshold as an additional input. Each sampled station-time example is expanded to the seven training thresholds. Station IDs are one-hot encoded and numeric inputs scaled. L2 regularisation is selected from 0.01 and 0.1 using 2022 mean Brier. Each fit allows up to 100 optimisation iterations and can stop earlier when the

convergence criterion is met. LR provides a relatively simple probability model against which to assess the tree methods; its probabilities receive no separate calibration fit.

**Threshold-conditioned gradient-boosted trees (GBT).** GBT uses the same binary labels and threshold expansion as LR, but learns nonlinear relationships by fitting trees sequentially. Its pipeline indexes station IDs, imputes numeric features and assembles them with $q$. We compare depths 3 and 5 with 40 boosting iterations, step size 0.1, minimum leaf size 25, subsampling 0.8 and 128 feature bins. Every reported configuration selects depth 5. Temperature calibration is applied to the binary probability pair $(1-p, p)$: A selects $T = 0.75$ and B selects $T = 1$ at each fraction. Unlike the RF survival mixture, this construction does not enforce decreasing probability as $q$ increases.

**Historical frequency.** This model estimates the event frequency for each station/month/hour group. We shrink each estimate towards the global training rate using 100 pseudo-observations and use that global rate for unseen groups. The baseline captures seasonal and daily patterns without fitting a classifier, making it a useful test of whether more complex models add value, especially in setting B.

# 5 Prediction results

## 5.1 Does recent rainfall improve the forecast?

Our main score is the Brier score,

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{p}_i - z_i)^2, \qquad z_i = \mathbf{1}(Y_i > q),$$

which measures probability squared error; lower is better. PR-AUC summarises the precision–recall tradeoff, while ROC-AUC measures ranking across true- and false-positive rates. We also report log loss, which penalises confident errors, and reliability, which compares predicted probabilities with observed frequencies. Spark computes AUC using 1,000-bin curve downsampling, and its PR-AUC is not average precision [5].

We first compare the 15% RFs on the same 5,729,508 examples, covering 89.9% of B's eligible 2024 set. Each forest uses the training population defined by its information requirements. Scoring both on shared test examples gives a paired comparison of the resulting forecasts.

| $q$ (mm) | Event rate (%) | Brier A | Brier B | PR-AUC A | PR-AUC B |
|---|---|---|---|---|---|
| 0 | 6.398 | 0.03798 | 0.05924 | 0.6091 | 0.1097 |
| 0.2 | 4.892 | 0.03090 | 0.04614 | 0.5582 | 0.0881 |
| 1 | 2.579 | 0.01870 | 0.02499 | 0.4215 | 0.0484 |
| 2 | 1.686 | 0.01349 | 0.01652 | 0.3155 | 0.0329 |
| 5 | 0.829 | 0.00732 | 0.00820 | 0.1885 | 0.0176 |
| 10 | 0.378 | 0.00354 | 0.00376 | 0.1020 | 0.0085 |
| 20 | 0.100 | 0.00098 | 0.00100 | 0.0304 | 0.0029 |

**Table 6.** The 15% RFs evaluated on identical 2024 examples. The event rate gives context for rare-event scores: a rare exceedance event can have a small Brier score even with little predictive information.

Across the seven thresholds, mean Brier falls from 0.02283 for B to 0.01613 for A, a 29.4% relative reduction in probability squared error. The advantage is greatest for any positive rain (35.9% reduction) and decreases to 1.7% above 20 mm. Recent rain is therefore useful in this holdout, but the heaviest events remain difficult.

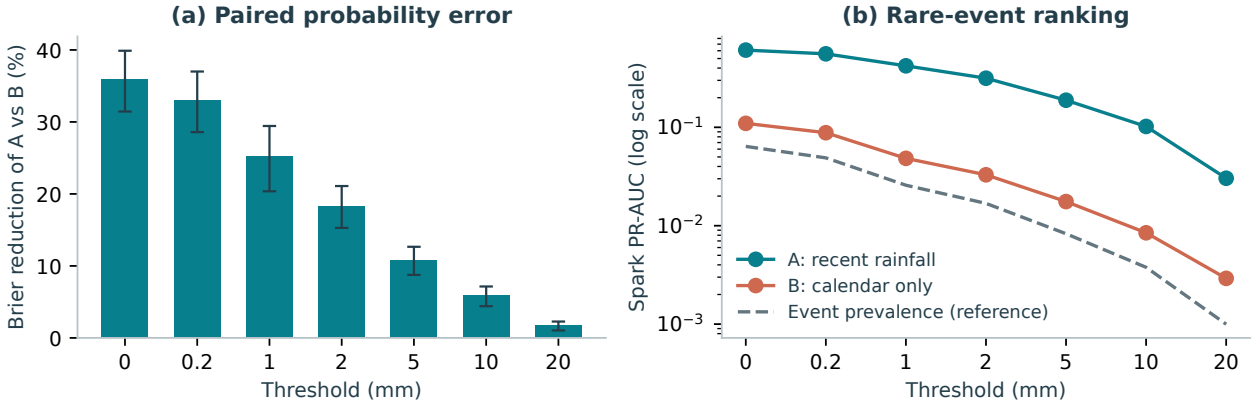



**Figure 9.** The 15% RF comparison on shared 2024 examples. Left: relative Brier reduction with conditional 95% block-bootstrap intervals. Right: PR-AUC and event prevalence on a logarithmic scale. Rainfall thresholds are displayed as equally spaced categories.

To account for overlapping labels and shared weather across stations, we form 2,000 bootstrap replicates by resampling circular blocks of seven consecutive calendar days, retaining all stations and thresholds together. The 95% interval for the mean relative Brier reduction is 25.4–32.9%, conditional on these fitted models and the observed 2024 network.

## 5.2 How do the model families compare?

| Mode | Model | Training | Mean Brier | PR-AUC at 1 mm | Log loss at 1 mm |
|---|---|---|---|---|---|
| A | Random forest | 100% | 0.01614 | 0.4229 | 0.0824 |
| A | Logistic regression | 100% | 0.01944 | 0.3634 | 0.0875 |
| A | GBT | 100% | 0.01623 | 0.4282 | 0.0824 |
| A | Historical frequency | 100% | 0.02270 | 0.0467 | 0.1166 |
| B | Random forest | 100% | 0.02275 | 0.0518 | 0.1167 |
| B | Logistic regression | 100% | 0.02301 | 0.0428 | 0.1165 |
| B | GBT | 100% | 0.02288 | 0.0522 | 0.1194 |
| B | Historical frequency | 100% | 0.02252 | 0.0516 | 0.1146 |

**Table 7.** Model comparison on each mode's eligible 2024 population: 5,729,508 examples for A and 6,374,476 for B. Every classifier in this table uses 100% of its eligible training population, so the algorithm comparison holds fitting fraction fixed. Historical frequency also uses the full training population. Mean Brier averages the seven original thresholds. Table 6 provides the comparison on shared A/B examples.

With every classifier fitted on the full training population, RF has the lowest mean Brier in A. GBT has slightly higher PR-AUC at 1 mm (0.4282 versus 0.4229), but higher mean probability squared error. LR has higher ROC-AUC at 1 mm (0.8611 versus 0.8036), while RF has better PR-AUC, Brier and log loss. Ranking events and assigning accurate probabilities therefore describe different aspects of performance.

For B, historical frequency achieves the lowest mean Brier. GBT has slightly higher PR-AUC at 1 mm than this baseline, but its probabilities are less accurate overall. Station and calendar features describe when rain is more likely on average; they contain much less information about the development of an individual storm than recent observations do.

## 5.3 How much training data is useful?

| Mode | Model | 15% | 30% | 100% | Improvement (%) |
|---|---|---|---|---|---|
| A | Random forest | 0.01613051 | 0.01613151 | 0.01613638 | -0.036 |
| A | Logistic regression | 0.01944150 | 0.01943988 | 0.01944300 | -0.008 |
| A | GBT | 0.01622929 | 0.01623889 | 0.01622724 | +0.013 |
| B | Random forest | 0.02274285 | 0.02274079 | 0.02275362 | -0.047 |
| B | Logistic regression | 0.02301006 | 0.02301067 | 0.02301076 | -0.003 |
| B | GBT | 0.02290321 | 0.02290187 | 0.02288430 | +0.083 |

**Table 8.** Mean Brier on each mode's eligible 2024 population, averaged over the seven original thresholds. Improvement is the relative Brier reduction from 15% to 100%; positive means lower error. All nine algorithm/fraction combinations are evaluated on 2024 in each information setting. The 15% A GBT mean comes from the retained aggregate summary; other cells use metric exports.

Using all eligible training examples changes mean Brier only slightly. For RF, the 100% score is 0.036% higher in A and 0.047% higher in B than at 15%. The corresponding changes for logistic regression are 0.008% and 0.003%. GBT's mean Brier falls by approximately 0.013% in A and 0.083% in B. A's 30% GBT score is slightly higher than its 15% score, so the observed learning curve is not monotonic. Across the three fitting fractions, RF has lower mean Brier than LR and GBT in both information settings. The separately fitted 30% LR changes mean Brier relative to 15% by +0.008% for A and -0.003% for B (positive denotes improvement). These values come from new LR fits, rather than the 15% references rescored in the GBT exports. At 30%, selected L2 regularisation is 0.01 for A and 0.01 for B; the selected fits use 25 and 10 iterations, respectively. The 30% RF changes mean Brier relative to 15% by -0.006% in A and +0.009% in B. Its existing tree weights are unchanged; probability calibration uses 2023, and these scores use the same eligible 2024 populations as the other fractions.

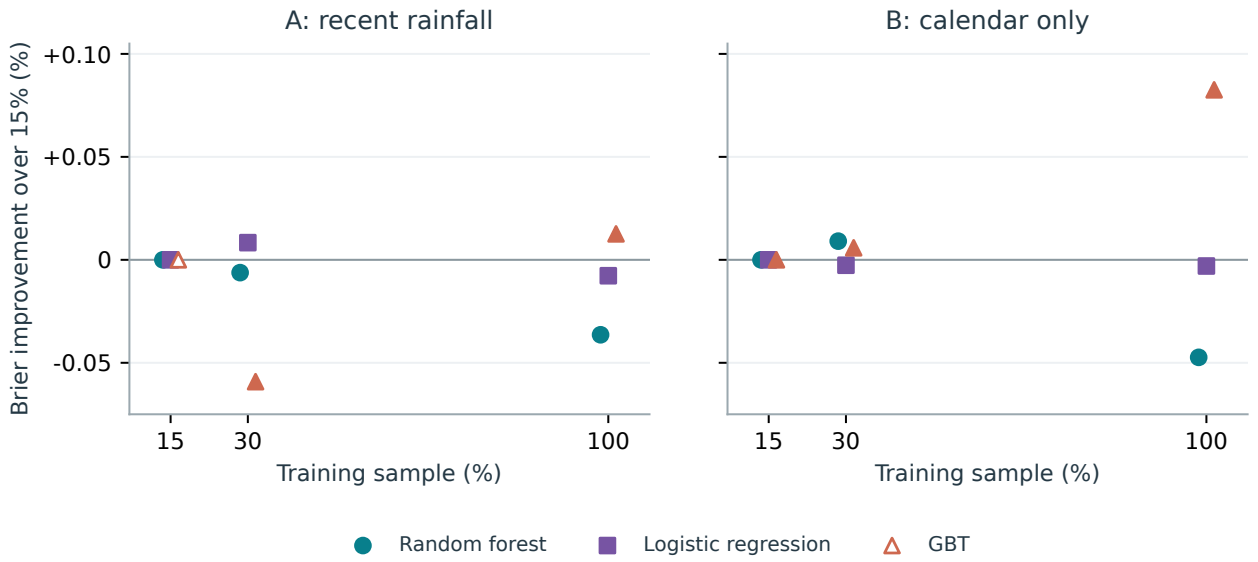



**Figure 10.** Relative improvement in mean 2024 Brier over each model's 15% result; positive values mean lower error. Points show the recorded fitting fractions, with small horizontal offsets for readability. All three algorithms have points at 15%, 30% and 100%. The open A GBT marker identifies the aggregate-summary reference.

The controlled RF comparison on 2022 provides a complementary check. With fitting settings held fixed, increasing the sample from 15% to 30% changes mean Brier by only 0.012% in A's favour and 0.003% against B; both conditional intervals include zero. A modest increase in tree depth is more useful for A than doubling the sample at the reference settings. Appendix G gives the individual settings and paired intervals. This validation analysis explains the tuning choices; the primary sample-size comparison is the complete 2024 grid in Table 8.

The 15% GBT scores come from an aggregate summary for A and per-threshold metrics for B. Appendix F lists the retained results and model packages.

### 5.4 Probability calibration and alarm decisions

A's RF distinguishes a wider range of risks, but several high-probability bins underestimate the event frequency. Mean predicted risk is 2.38%, against 2.58% observed, and the ten-bin weighted absolute calibration error is 0.196 percentage points. These figures use A's full eligible set, which is also the shared set. Most examples occupy low-probability bins, so the small aggregate error should be read alongside the sample counts and the deviations at higher risks.

At the matched 30% fitting fraction, GBT's mean predicted risk above 20 mm is 0.830% in A and 2.645% in B, compared with observed frequencies of 0.100% and 0.103%. This overprediction helps explain why event ranking and probability error can favour different models. Temperature calibration uses aggregate Brier on 2023; this plot checks threshold-specific behaviour on 2024.

Finally, a rainfall threshold and an alarm cutoff are different choices. At $q = 1$ mm, changing A's probability cutoff from 0.5 to the validation-selected 0.2 raises recall from 24.7% to 46.4%, while precision falls from 69.3% to 47.4%. Cutoffs are selected by 2022 F1 after fixing the 2023 temperature, without using 2024 outcomes. An always-negative rule already achieves about 97.4% accuracy here, which is why probability quality and the precision–recall tradeoff are more informative than accuracy.

### 5.5 Queries between the training thresholds

The additional queries at 3, 7.5 and 15 mm test behaviour inside the RF rainfall bins and between the thresholds used to fit LR and GBT. Table 9 compares RF, LR and GBT at a matched 30% fitting fraction on each mode's complete eligible 2024 population.

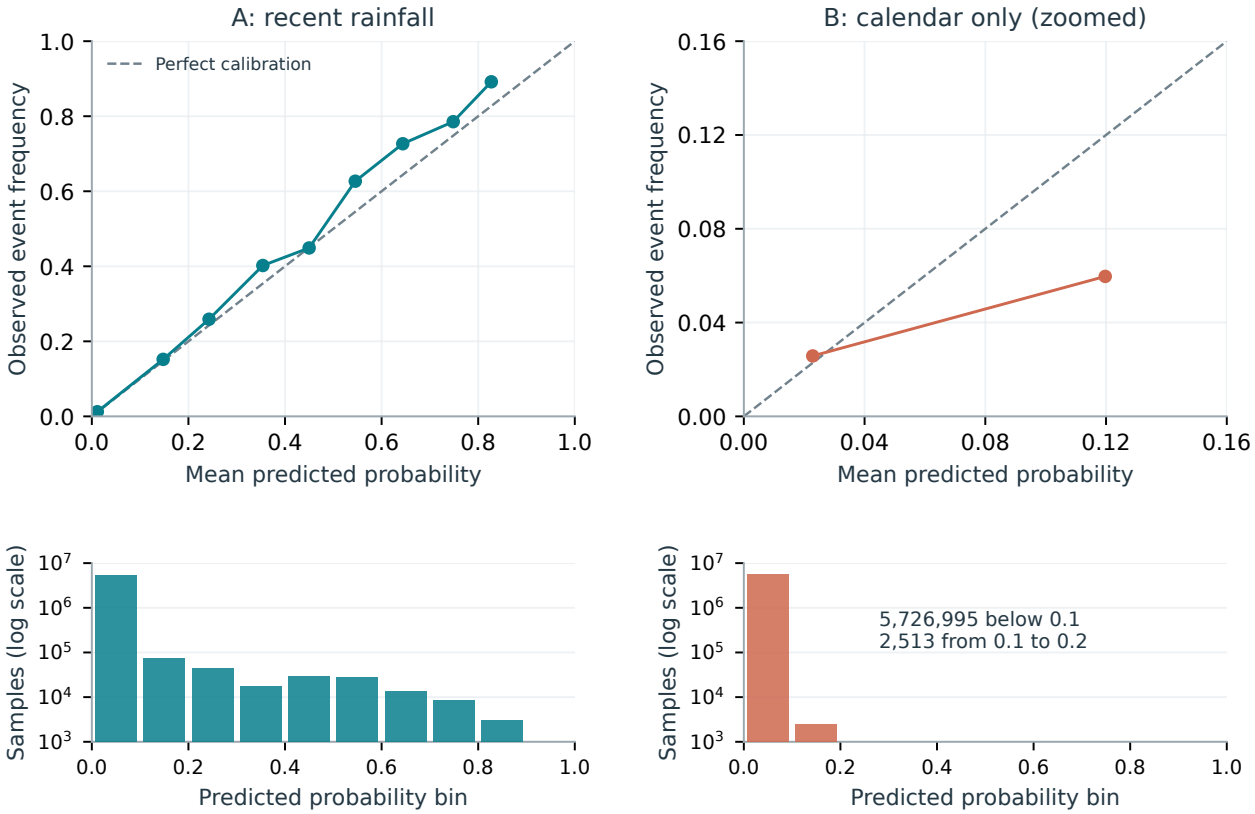



**Figure 11.** Calibration of the 15% RFs at 1 mm on shared examples. Upper panels compare bin-average predictions with observed event rates; B is enlarged to show its two occupied bins. Lower panels show sample counts in fixed 0.1-wide probability bins across the full probability range.

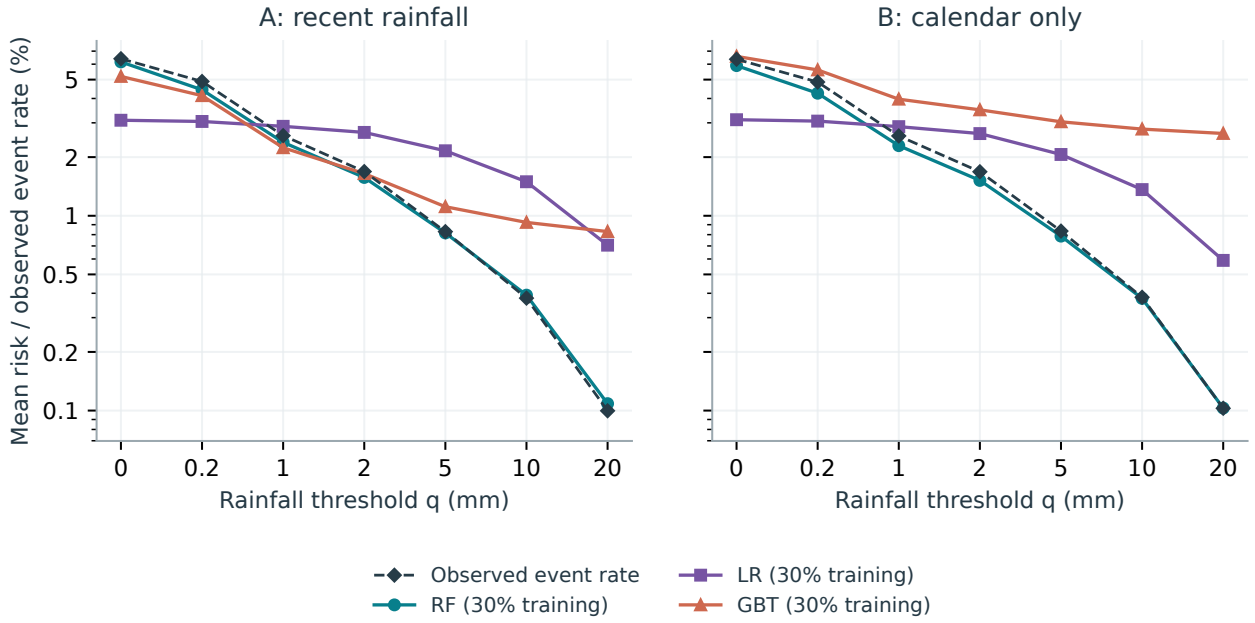



**Figure 12.** Observed event frequencies and mean predicted probabilities from RF, LR and GBT, each fitted on 30% of training anchors, at the seven original thresholds. Each panel uses its mode's eligible 2024 examples. The vertical axis is logarithmic and thresholds are equally spaced categories. Agreement in the mean is one calibration check; the reliability plot examines variation within probability groups.

| Mode | $q$ (mm) | RF Brier | LR Brier | GBT Brier |
|---|---|---|---|---|
| A | 3 | 0.010546 | 0.012359 | 0.010717 |
| A | 7.5 | 0.005051 | 0.006950 | 0.005132 |
| A | 15 | 0.001854 | 0.003118 | 0.001913 |
| B | 3 | 0.012385 | 0.012510 | 0.012856 |
| B | 7.5 | 0.005534 | 0.005663 | 0.006142 |
| B | 15 | 0.001950 | 0.002002 | 0.002620 |

**Table 9.** 2024 results at thresholds absent from the fitting query grid. All three algorithms use 30% of training anchors and the same eligible test population within each mode.

RF has lower Brier than both alternatives in all six matched-fraction queries. Its global within-bin approximation therefore remains useful at these additional query points. GBT also exposes a structural issue with treating the threshold as an unconstrained predictor: a larger rainfall threshold can occasionally produce a larger exceedance probability. Across the ten evaluated thresholds, the proportion of adjacent query pairs with increasing GBT probability is 0.0491% at 30% training and 0.0681% at 100% for A, compared with 0.5238% and 0.6164% for B. The RF mixture preserves the expected probability ordering across thresholds by construction.

Appendix E retains threshold-specific baseline skill, monthly skill and feature-importance diagnostics for the 15% models. The repeated AUC values across tables can differ slightly because Spark downsamples the curves according to their partitions.

## 6 Discussion

The largest predictive difference comes from the information available at issue time. Recent rainfall describes the persistence and development of local rain, while calendar features describe recurring seasonal and daily patterns. This explains why A is useful for short-term forecasting and why a historical-frequency table remains competitive in B. The weather exploration and the model comparison answer complementary questions: when rain is common across years, and how current observations change the risk for a particular half-hour.

The additional training experiments sharpen the model choice. Across the complete 15%, 30% and 100% comparison on 2024, RF has the lowest mean Brier among the classifiers, while fitting every eligible training example changes its score very little. At 100% training, GBT's slightly higher PR-AUC at 1 mm shows useful ranking ability, but its average predicted risk at 10 and 20 mm is too high. Model development should therefore focus on the probability estimates for large accumulations as well as on ranking events. Threshold-specific calibration and a monotonic threshold-conditioned model are natural extensions of the present comparison.

The sampling results also help allocate computation. The 15% sample already contains millions of training anchors. Adjacent anchors share rainfall observations because they are only five minutes apart, so a larger row count includes substantial repeated information about the same weather episodes. In the RF validation study, allowing deeper trees produces a larger improvement than doubling the reference model's sample. Together, these findings favour testing model settings and tail behaviour before committing to larger fits of the same configuration.

Coverage remains part of the population being described. The exploration retains 85 of 96 year-months, and A's history requirement retains 89.9% of B's eligible 2024 cases. Its performance therefore describes forecast times with complete published history. Extending the evaluation across successive years and grouping results by season and station would show how stable the findings are as weather and network coverage change.

Operational use requires recent observations and an up-to-date station catalogue. The current implementation treats observation times as interval endpoints and checks both publication timestamps. A live archive would record what was actually available at each forecast time, while catalogue updates would track station relocations. These additions would connect the offline experiment to routine forecasting.

## 7 Conclusion

The records show seasonal, diurnal and local variation across Singapore. Coverage-aware monthly summaries and common-station comparisons make these patterns visible despite gaps in the annual files. For prediction, recent-rainfall random forests provide the strongest overall probability estimates among the tested classifiers. Larger training fractions give small changes, while GBT adds competitive ranking and a clear need to improve high-threshold calibration. The resulting model choice is a recent-rainfall forest for short-term queries, with historical frequency the strongest calendar-risk baseline at the evaluated thresholds.

## AI usage declaration

OpenAI Codex assisted with code development, data-cleaning diagnostics, cloud execution, feature and model implementation, tests, figure preparation, numerical analysis and report drafting. It also assisted with restructuring the report, integrating the RF/LR/GBT experiments and producing figures from saved results. Reported numerical results come from the recorded data analyses and model outputs; synthetic records were used for software tests. Group members remain responsible for checking the submitted code, figures, interpretations and references.

## References

[1] National Environment Agency. *Historical Rainfall across Singapore*, annual datasets 2017–2024. Data.gov.sg. https://data.gov.sg/collections/2279/view. Accessed 9 October 2026.

[2] Meteorological Service Singapore. *Climate of Singapore*. https://www.weather.gov.sg/climate-climate-of-singapore/. Accessed 9 October 2026.

[3] geoBoundaries / Urban Redevelopment Authority. Singapore outline and planning-area geometries. Source and licence records are retained in the project assets; planning-area geometry is based on URA Master Plan 2014.

[4] Apache Spark. *Classification and regression*, Spark 3.5.3 documentation. https://dlcdn.apache.org/spark/docs/3.5.3/ml-classification-regression.html. Accessed 9 October 2026.

[5] Apache Spark. *BinaryClassificationMetrics*, Spark 3.5.3 Java API. https://spark.apache.org/docs/3.5.3/api/java/org/apache/spark/mllib/evaluation/BinaryClassificationMetrics.html. Accessed 9 October 2026.

## A Distribution and source-gap evidence

These supplementary figures separate the distribution of measured rain from the absence of measurements. They support the data-quality discussion without treating an unrecorded interval as dry weather.

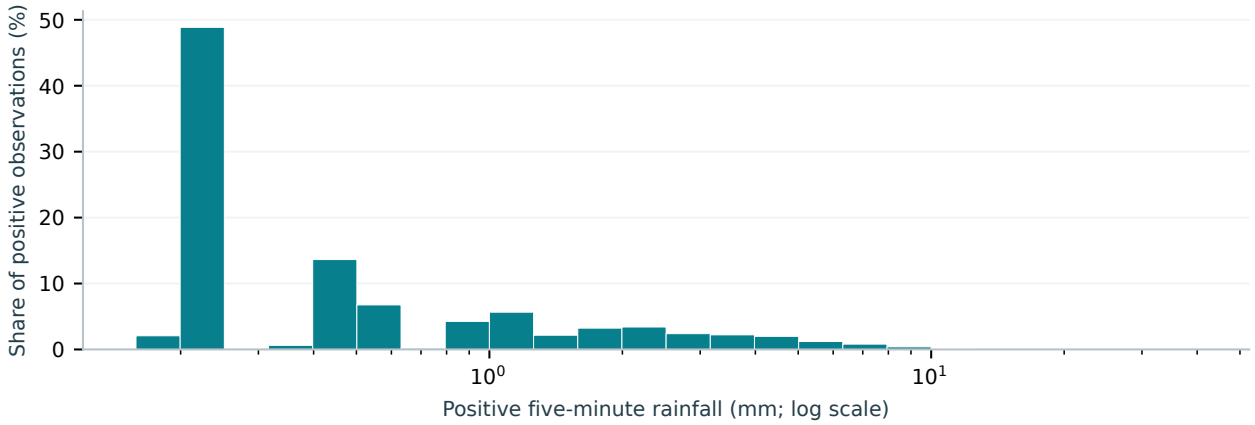



**Figure A.1.** Distribution of positive five-minute rainfall across 2017–2024. Bars give the share of positive observations in log-spaced bins; the horizontal axis is logarithmic. Recorded zeros are excluded. The long upper tail motivates evaluating rare accumulations separately.

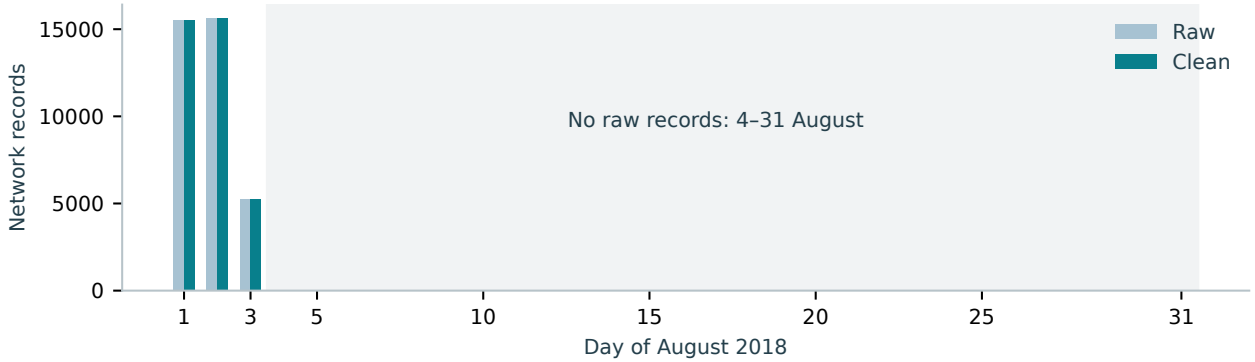



**Figure A.2.** Daily network-wide raw and cleaned record counts in August 2018. The absence of observations on 4–31 August is already present in the source file. The cleaning process therefore does not explain this gap, and this figure does not identify its upstream cause.

## B Monthly trajectories and annual support

The monthly trajectories retain usable periods individually. The annual plot applies the stricter full-year rule, showing why monthly evidence is more extensive than annual evidence.

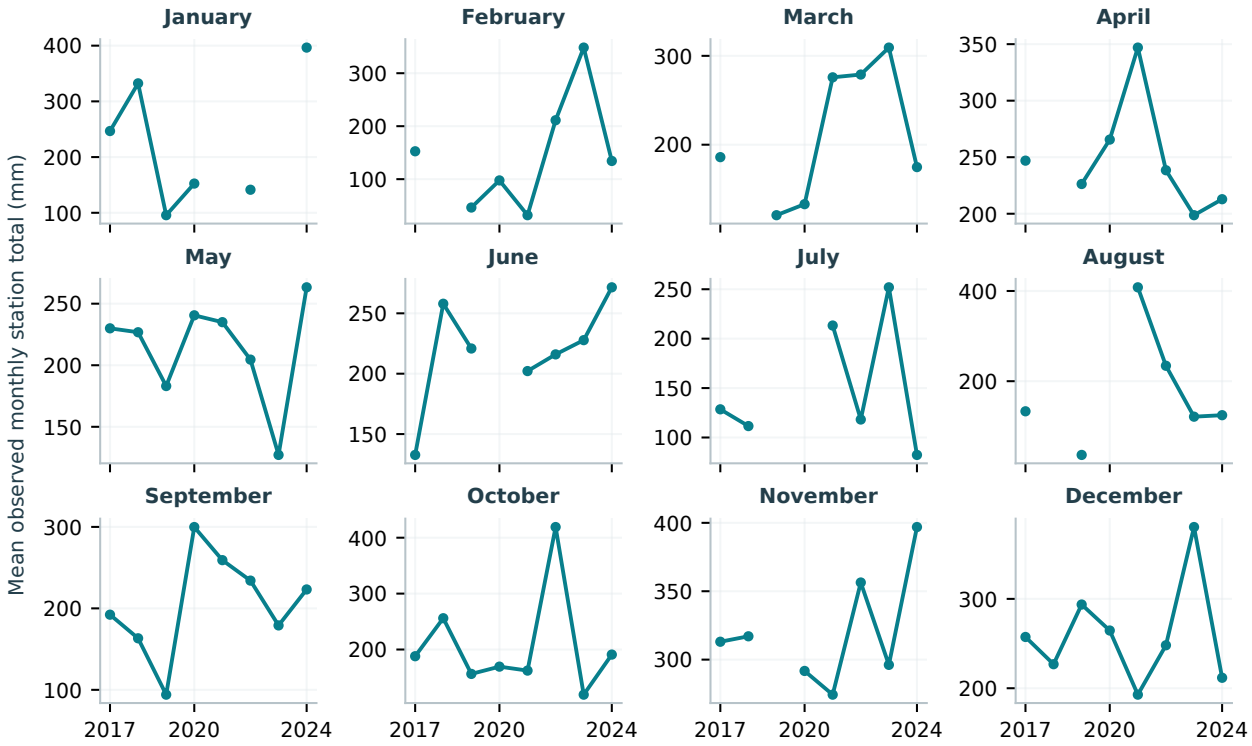



**Figure B.1.** Mean observed monthly station totals, with a separate trajectory for each calendar month. Missing year-months break the lines. Vertical ranges differ across panels; use the main monthly heatmap to compare absolute amounts between months. Station composition can change between points, so these are descriptive network trajectories.

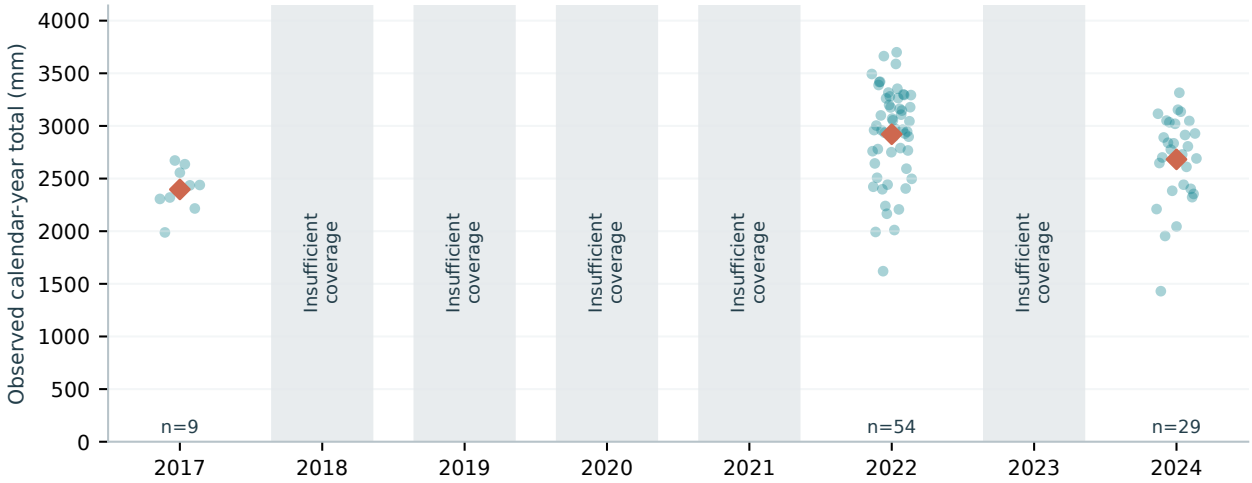



**Figure B.2.** Observed calendar-year totals at eligible stations. Points show station totals, coral diamonds their mean, and $n$ the contributing station count. Grey columns have no station meeting annual coverage of at least 95% and coverage of at least 90% in every month. The three supported years use different station sets and do not establish a national wettest-year ranking.

## C Daily rainfall and coverage sensitivity

Daily upper-tail amounts and wet-day frequencies describe different aspects of rainfall. The sensitivity curves show how changing the completeness rule affects seasonal summaries, while retaining the limitations of the observed network.

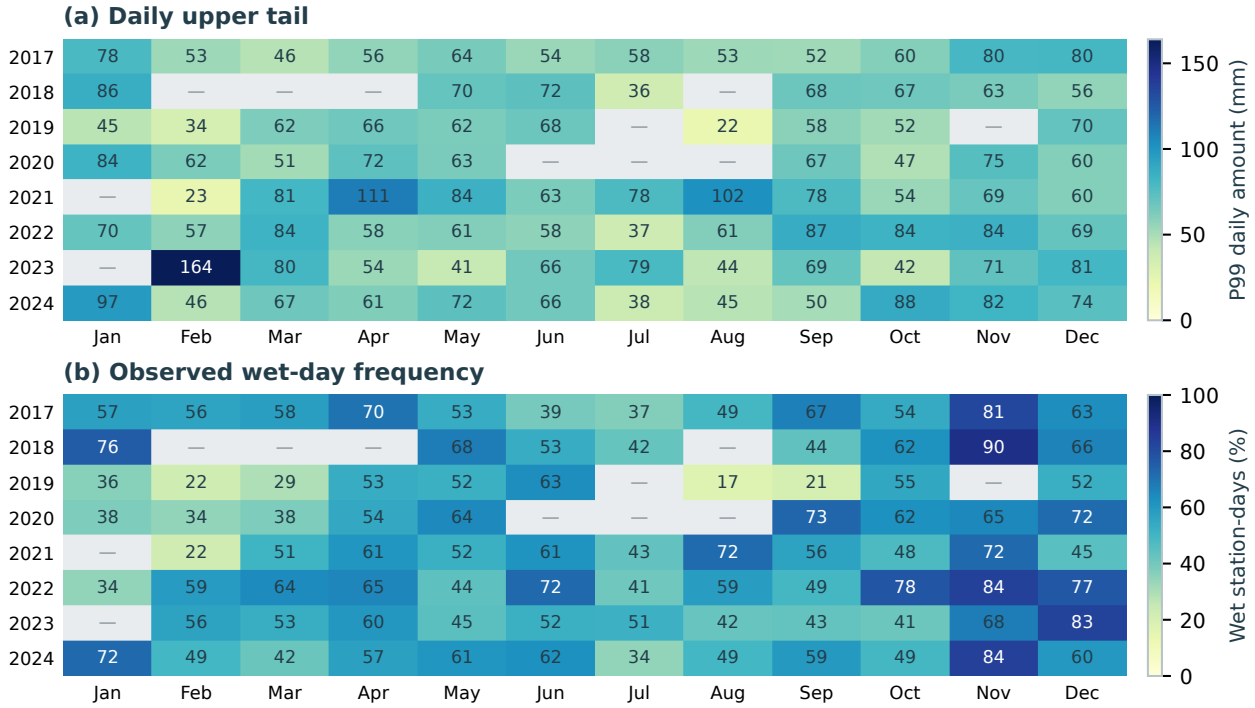



**Figure C.1.** Monthly daily-rainfall P99 (upper) and wet station-day percentage (lower). Included days have at least 95% daily coverage within eligible station-months. Quantiles include dry days; wet means observed daily rainfall above zero. P99 is an approximate Spark quantile of station-days, not an independent-storm return level. Grey cells lack qualifying support.

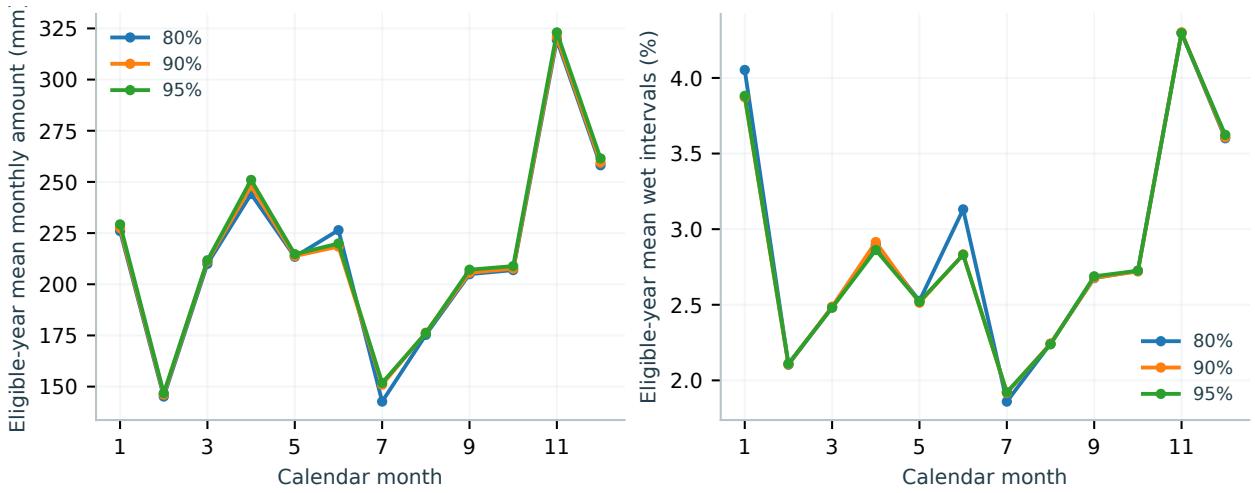



**Figure C.2.** Seasonal mean amount and wet-interval frequency under 80%, 90% and 95% monthly coverage cutoffs. Each curve uses its own eligible stations and equally weighted available years. Similar curves support the stability of these summaries under the tested cutoffs; they do not establish random missingness.

## D Additional spatial views

The twelve-month view complements the main December comparison. Regional colours summarise contributing gauges and should be read together with their locations and coverage.




**Figure D.1.** All twelve months of 2024. Colours show observed monthly totals at qualified stations, and $n$ gives their number. The original gallery's global monthly colour scale is retained; it differs from the December-only scale in the main text. The eligible network changes by month, and no rainfall is interpolated between gauges.

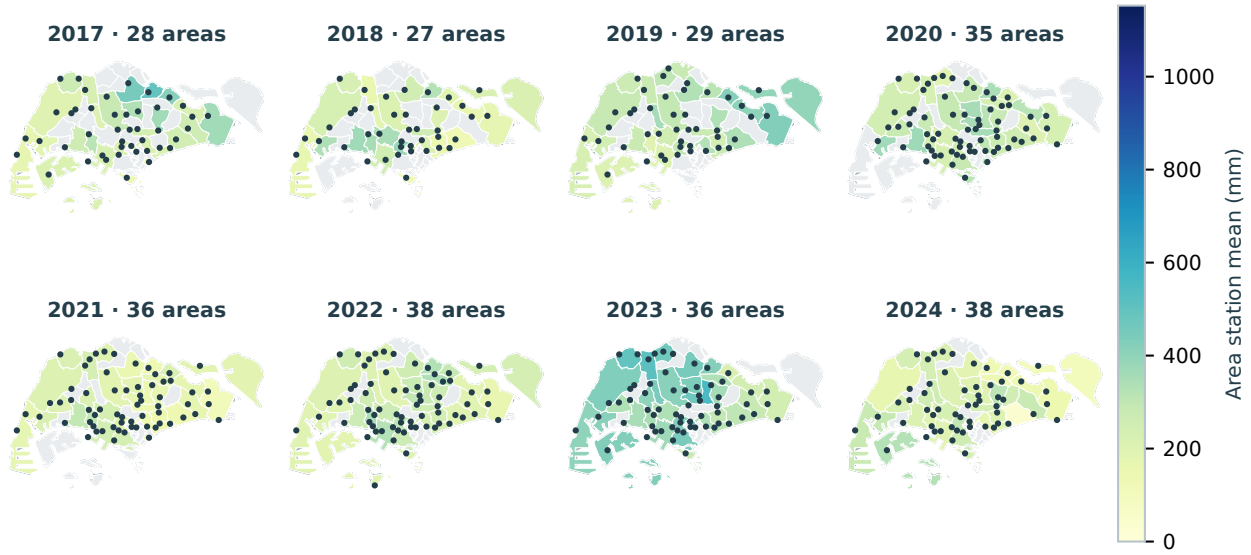



**Figure D.2.** December planning-area station means across eight years. Each coloured polygon shows the arithmetic mean of its assigned eligible stations; dots show contributing locations. Grey means no supporting station, not zero rainfall. The original regional scale is retained. These are gauge summaries, not area-weighted rainfall estimates; station counts and placements affect their representativeness.

## E Additional predictive diagnostics

These figures describe the 15% RF and logistic models on the 2024 test set. They show how performance varies by rainfall threshold and month, alongside the RF's feature use.

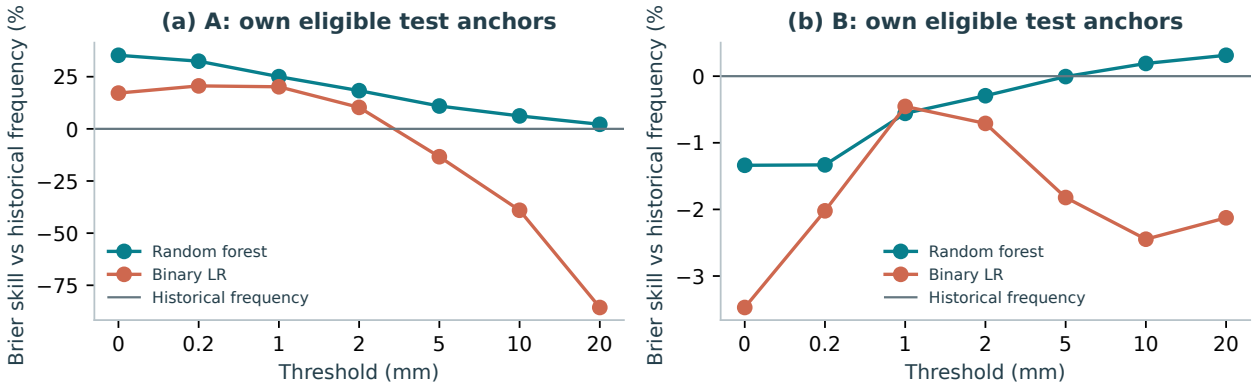



**Figure E.1.** Brier skill relative to each mode's historical-frequency baseline on its own eligible 2024 test population. Positive values indicate lower error than the baseline; zero indicates equal error. A's random forest improves on the baseline at every evaluated threshold, whereas A's logistic regression deteriorates at larger thresholds. B's forest has small, threshold-dependent gains and losses. The two panels use different eligible populations.

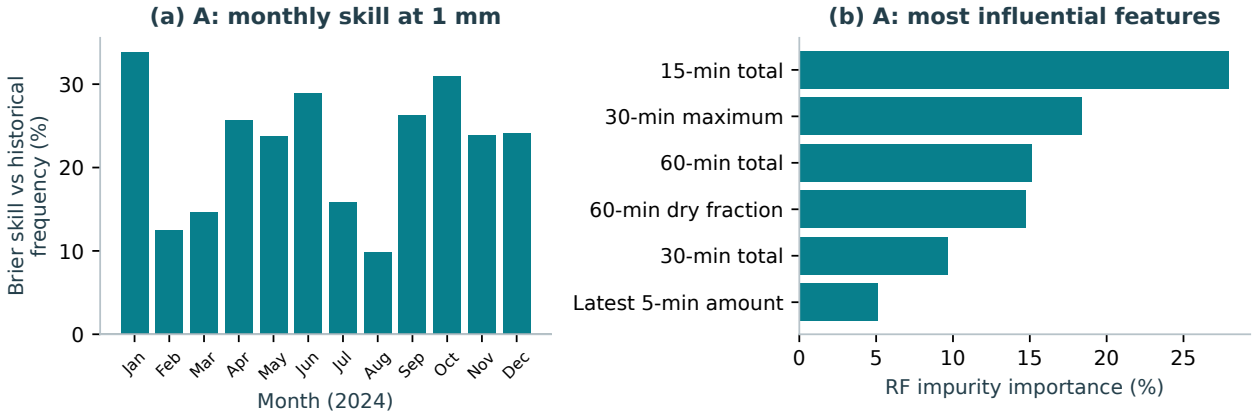



**Figure E.2.** Left: A's monthly Brier skill relative to historical frequency at 1 mm in 2024. Right: its six largest random-forest impurity importances. Recent rainfall totals and persistence summaries contribute most to the fitted forest; these correlated features share information.

## F Model scores and experiment coverage

Table F.1 brings the recorded model and training-fraction results together. Mean Brier and mean log loss give equal weight to the seven primary thresholds; PR-AUC is shown at 1 mm. Historical frequency uses all eligible training examples in every run.

| Mode | Model | Training | Mean Brier | Mean log loss | PR-AUC at 1 mm |
|---|---|---|---|---|---|
| A | Random forest | 15% | 0.01613051 | 0.070004 | 0.4218 |
| A | Random forest | 30% | 0.01613151 | 0.070021 | 0.4225 |
| A | Random forest | 100% | 0.01613638 | 0.070016 | 0.4229 |
| A | Logistic regression | 15% | 0.01944150 | 0.082787 | 0.3629 |
| A | Logistic regression | 30% | 0.01943988 | 0.082783 | 0.3632 |
| A | Logistic regression | 100% | 0.01944300 | 0.082788 | 0.3634 |
| A | GBT | 15% | 0.01622929 | 0.072167 | — |
| A | GBT | 30% | 0.01623889 | 0.072197 | 0.4203 |
| A | GBT | 100% | 0.01622724 | 0.072151 | 0.4282 |
| A | Historical frequency | 100% | 0.02269628 | 0.099923 | 0.0467 |
| B | Random forest | 15% | 0.02274285 | 0.100015 | 0.0494 |
| B | Random forest | 30% | 0.02274079 | 0.099983 | 0.0488 |
| B | Random forest | 100% | 0.02275362 | 0.100170 | 0.0518 |
| B | Logistic regression | 15% | 0.02301006 | 0.104528 | 0.0428 |
| B | Logistic regression | 30% | 0.02301067 | 0.104539 | 0.0428 |
| B | Logistic regression | 100% | 0.02301076 | 0.104540 | 0.0428 |
| B | GBT | 15% | 0.02290321 | 0.107891 | 0.0511 |
| B | GBT | 30% | 0.02290187 | 0.107892 | 0.0521 |
| B | GBT | 100% | 0.02288430 | 0.107829 | 0.0522 |
| B | Historical frequency | 100% | 0.02252051 | 0.098657 | 0.0516 |

**Table F.1.** 2024 results on each mode's eligible population: 5,729,508 anchors for A and 6,374,476 for B. The 15% A GBT row uses its recorded aggregate summary; the 15% B GBT row uses per-threshold exports. All other rows use saved metric CSVs.

The RF, LR and GBT comparison contains all three fitting fractions on 2024. RF 30% uses the standard depth-8, 40-tree pipelines from the earlier sensitivity study, with a separate 2023 calibration and 2024 evaluation. LR 30% was fitted in a later LR-only run. RF/LR 15% and 100%, RF/LR 30%, and GBT 30% and 100% retain model pipelines and metric exports. The 15% A GBT row is supported only by its rounded aggregate summary; B's 15% GBT row has per-threshold text exports. These two 15% GBT pipelines were not supplied. Within GBT exports, RF/LR columns are scores of the loaded 15% reference models and do not represent additional RF/LR fits. The repository's Part 2 guide maps every result to its source directory.

| Execution batch | Fitted or reused models | Recorded evaluation |
|---|---|---|
| Initial experiment | RF and LR at 15%, A and B | 2022 selection; 2024 test |
| RF sensitivity | RF at 15% and 30%; one-factor tuning | 2022 validation |
| Full training | RF and LR at 100%, A and B | 2022 selection; 2024 test |
| GBT experiment | GBT at 15%, 30%, 100%, A and B | 2023 calibration; 2024 test |
| LR 30% completion | Newly fitted LR at 30%, A and B | 2022 selection; 2024 test |
| RF 30% completion | Reused standard RF 30% weights, A and B | 2023 calibration; 2024 test |

**Table F.2.** Execution provenance for the unified comparison. All classifier fitting uses 2017–2021. Later batches reuse 2024 after the initial results were available; they are not a new independent test year.

## G Random-forest tuning on 2022 validation

This experiment varies the training fraction and RF settings on the 2022 validation population. The reference forest uses 40 trees, depth 8, minimum leaf size 25, square-root feature selection and subsampling of 0.8. A deterministic station–time hash with seed 2026 gives nested 15% and 30% samples. The 30% samples contain 5,608,769 anchors for A and 8,233,882 for B. Every candidate within a mode uses the same validation anchors, training-derived within-bin distributions and temperature $T = 1$.

At 15%, the one-factor comparisons increase depth to 10, increase the number of trees to 80, increase minimum leaf size to 50, or use half the features at each split. The lowest-Brier configuration at 15% is then fitted at 30%. Table G.1 gives all candidate scores.

| Configuration | Fitting sample | A mean Brier | B mean Brier |
|---|---|---|---|
| Reference | 15% | 0.01669750 | 0.02332733 |
| Reference | 30% | 0.01669544 | 0.02332800 |
| Depth 10 | 15% | 0.01664745 | 0.02331832 |
| 80 trees | 15% | 0.01669631 | 0.02332818 |
| Minimum leaf size 50 | 15% | 0.01671066 | 0.02333420 |
| Half-feature selection | 15% | 0.01668206 | 0.02331674 |
| Selected setting | 30% | 0.01663765 | 0.02330797 |

**Table G.1.** Mean Brier on the full 2022 validation set, averaged over seven rainfall thresholds. Lower is better. Changes at 15% vary one setting from the reference. The selected 30% row uses depth 10 for A and half-feature selection for B; it is not the same configuration for both modes.

At the reference settings, increasing the sample to 30% reduces A's mean Brier from 0.01669750 to 0.01669544, a relative improvement of 0.012%. B's mean Brier increases from 0.02332733 to 0.02332800, a relative deterioration of 0.003%. The corresponding 95% intervals for improvement are $[-0.014\%,\ 0.039\%]$ for A and $[-0.070\%,\ 0.047\%]$ for B.

Depth has a larger effect for A. Moving from depth 8 to 10 at 15% improves mean Brier by 0.300%, with interval $[0.213\%,\ 0.389\%]$. Fitting that depth-10 configuration at 30% adds a further 0.059% improvement, with interval $[0.036\%,\ 0.082\%]$, and reaches 0.01663765. B's best tested 15% setting uses half the features; its 30% version reaches 0.02330797. Both B intervals for the configuration and subsequent sample changes cross zero. At the reference settings, A's Brier improves by 0.103% at 10 mm and 0.072% at 20 mm.

Intervals use 2,000 paired circular resamples of seven-day blocks, retaining all stations and thresholds together. They describe validation-day variation for the fitted models and seed used here. Configuration selection uses the same validation year. These Brier comparisons complement the 2024 model and sample-size results in the main text.